
CISC/CMPE 452/COGS 400

Assignment 1 Theoretical Part

1. **Perceptron:** - Consider the classification problem defined below.
 $\mathbf{p}_1 = ([-1, 1]^t, 1)$, $\mathbf{p}_2 = ([-1, -1]^t, 1)$, $\mathbf{p}_3 = ([0, 0]^t, 0)$, and $\mathbf{p}_4 = ([1, 0]^t, 0)$.
 - (a) Design a single-neuron perceptron to solve this problem. Design the network graphically, by choosing weight vectors that are orthogonal to the decision boundaries.
 - (b) Test your solution with all four input vectors.
 - (c) Classify the following input vectors with your solution $\mathbf{p}_5 = ([-2, 0]^t)$, $\mathbf{p}_6 = ([1, 1]^t)$, $\mathbf{p}_7 = ([0, 1]^t)$, and $\mathbf{p}_8 = ([-1, -2]^t)$.
 - (d) Which of the vectors in part (3) will always be classified the same way, regardless of the solution? Which may vary depending on the solution? Why?
2. **Multilayer Perceptron** - You are presented with the following input set:
 $\mathbf{x}_1 = [3, 1, 1]^t$, $\mathbf{x}_2 = [4, 0, 1]^t$, $\mathbf{x}_3 = [4, -1, 1]^t$, $\mathbf{x}_4 = [5, 2, 1]^t$, $\mathbf{x}_5 = [5, 3, 1]^t$, $\mathbf{x}_6 = [3, 3, 1]^t$, $\mathbf{x}_7 = [2, 0, 1]^t$, and $\mathbf{x}_8 = [1, 1, 1]^t$. A neural network with two discrete bipolar perceptrons in the hidden layer and a single discrete bipolar output perceptron needs to classify the presented inputs in either of two classes, C_1 and C_2 such that $x_1, x_2, x_3 \in C_1$, with the remaining inputs belonging to C_2 .
 - i- Check whether the weights $\mathbf{w}_1 = [2, 1]^T$, $\theta_1 = 5$, and $\mathbf{w}_2 = [0, 1]^T$, $\theta_2 = -2$ would provide the linear separation of patterns as required.
 - ii- Repeat part i for the weights $\mathbf{w}_1 = [0, -1]^T$, $\theta_1 = 1.5$, and $\mathbf{w}_2 = [1, 0]^T$, $\theta_2 = -2.5$
 - iii- Complete the design of the classifier by using the results from either part i or ii and compute the weights of the single perceptron at the output.
3. **Constructing a Network** - Construct a multilayer perceptron which will be able to separate the two classes shown in the figure. Use two neurons in the output layer and find a suitable number of hidden layer neurons. Two output of the network should be $[1, 0]^t$ if the input belongs to class C_1 and $[0, 1]^t$ if the input belongs to class C_2 . Use the activation function

$$\sigma(\text{net}) = \begin{cases} 1 & \text{if } \text{net} > 0; \\ 0 & \text{if } \text{net} \leq 0. \end{cases}$$

and determine the weight by hand. What is the minimum amount of neurons in the hidden layer required for a perfect separation of the classes?

